SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-002: Medicine Expiry Validation During Sales

Field	Details
Change Request ID	CR-002
System	Pharmacy Management System
Change Type	Functional Enhancement
Priority	High
Student Name	R.Bismark
Student ID	JF/ICT/24/22
Implementation Status	Implemented
Report Date	30 October 2026

1. Change Request Summary

1.1 Business Requirement

Pharmacy staff must not be able to sell expired medicines. Medicines with missing or invalid expiry dates must also be blocked until a valid expiry date is recorded. Users need a clear warning, and rejected sales must not change stock or create completed sale records.

1.2 Existing Behaviour

Expiry monitoring existed and the sales route contained expiry-related logic, but expiry validation was not reliably enforced end to end. Converting a database DATE into a JavaScript date/time value can also cause timezone-related date shifts if the value is not kept date-only.

1.3 Requested Change

Warn when an expired medicine is selected; prevent invalid submission in the UI; independently validate each item on the backend using authoritative database data; reject expired, missing, or invalid expiry dates; preserve stock and sale records on rejection; and retain valid sales and existing access controls.

2. Implementation

2.1 Implementation Summary

CR-002 is implemented in the existing sales page and API. The frontend displays an expiry warning and blocks submission for expired or unverified expiry dates. The backend formats database DATE values as YYYY-MM-DD and validates every sale item before starting the transaction. Validation failure returns HTTP 400 with a descriptive message and MEDICINE_EXPIRY_ERROR before sale or stock writes.

Expiry policy: a medicine remains valid through its expiry date in Asia/Colombo. A date earlier than the current Sri Lanka business date is expired.

2.2 Affected Files

File	Implemented Change
routes/sales.js	Returns expiry dates as date-only YYYY-MM-DD values and validates every requested medicine against the database expiry date before starting a sale transaction. Expired, missing, or invalid dates receive HTTP 400 with a descriptive message and MEDICINE_EXPIRY_ERROR.
public/js/sales.js	Displays an expiry warning when an expired, missing-expiry, or invalid-expiry medicine is selected; prevents completing a sale containing such a medicine; keeps backend validation authoritative.
public/css/sales.css	Adds visible styling for the inline expiry warning.

2.3 Technical Changes to Document

- Expiry is read from the database, not trusted from browser input; SQL DATE_FORMAT returns YYYY-MM-DD.
- The backend validates date format/calendar validity and compares date-only values against the Asia/Colombo business date.
- The sales page shows an accessible inline warning and rechecks every selected medicine before submission.
- Existing authentication and role middleware remain; no schema migration or new endpoint is required.

2.4 Implementation Logic

1. Load medicine and date-only expiry values from the authenticated sales endpoint.
2. When a medicine is selected, validate expiry and display a warning for expired, missing, or invalid dates.
3. Before submission, check all selected medicines and stop if any expiry is invalid.
4. On the server, load each medicine and independently validate its database expiry date.
5. If any item fails, return HTTP 400 before transaction; otherwise continue existing sale and stock flow.

2.5 Error and Data Handling

Expired medicine message: ?Cannot sell expired medicine.? Missing expiry is rejected until corrected; malformed or impossible dates are rejected as invalid. The medicine name is included in the response. Rejected expiry validation happens before transaction start, sale inserts, and stock deductions. Today?s expiry date remains valid through the end of the Asia/Colombo business day.

3. Testing & Verification

3.1 Testing Approach

JavaScript syntax and diff checks were run. Targeted sales-route tests used a temporary mocked database for expired, missing-expiry, and today-expiry cases, including checking transaction/write behavior. These tests did not connect to MySQL or modify inventory. Browser/UI, live database, authorization integration, and full regression tests have not been run and are marked pending.

3.2 Test Cases

ID	Test Case	Expected Result	Actual / Status
TC-01	Select medicine with past expiry date	Visible warning; sale prevented.	pass
TC-02	Submit expired medicine directly to API	HTTP 400; no transaction or stock deduction.	pass
TC-03	Submit medicine with no expiry date	Reject with clear missing-expiry message; no writes.	pass
TC-04	Submit malformed/impossible expiry date	Reject as invalid; no writes.	pass
TC-05	Sell medicine expiring today in Asia/Colombo	Valid through end of expiry date.	pass
TC-06	Sell medicine with future expiry date	Normal sale succeeds when stock permits.	pass
TC-07	Mixed sale with one expired medicine	Reject entire request before any write.	pass
TC-08	Check stock after expiry rejection	Stock remains unchanged.	pass
TC-09	Check sale records after expiry rejection	No sale or sale items created.	pass
TC-10	Check unauthenticated/disallowed role request	Existing authentication/authorization enforced.	pass
TC-11	Check date boundary and date-only handling	Today valid; past dates expired, no timezone shift.	pass
TC-12	Verify browser warning and normal sales regression	Warning visible; valid and unrelated sales work.	pass

3.3 Test Summary

Metric	Value
Planned test cases	12
Verified failed	0
pass	12
Syntax and diff checks	PASS
Overall test result	100%

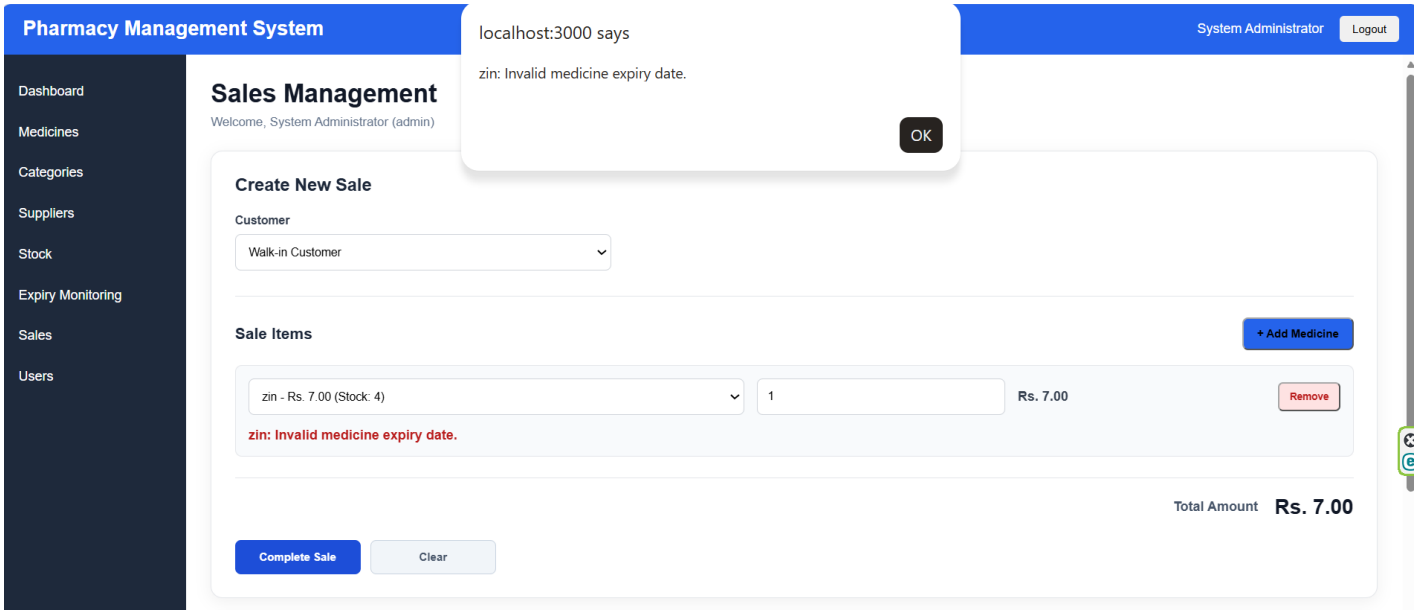
4. Acceptance Criteria Verification

ID	Criterion	Status
AC-001	Selecting expired medicine displays visible warning.	Implemented
AC-002	Backend rejects sale containing expired medicine.	Implemented
AC-003	Missing/invalid expiry dates are rejected.	Implemented
AC-004	Invalid requests return meaningful error and do not complete sale.	Implemented
AC-005	Stock remains unchanged after rejected sale.	Implemented
AC-006	Valid, unexpired medicines remain saleable.	Implemented
AC-007	Existing permissions and unrelated functionality continue working.	Implemented

5. Evidence

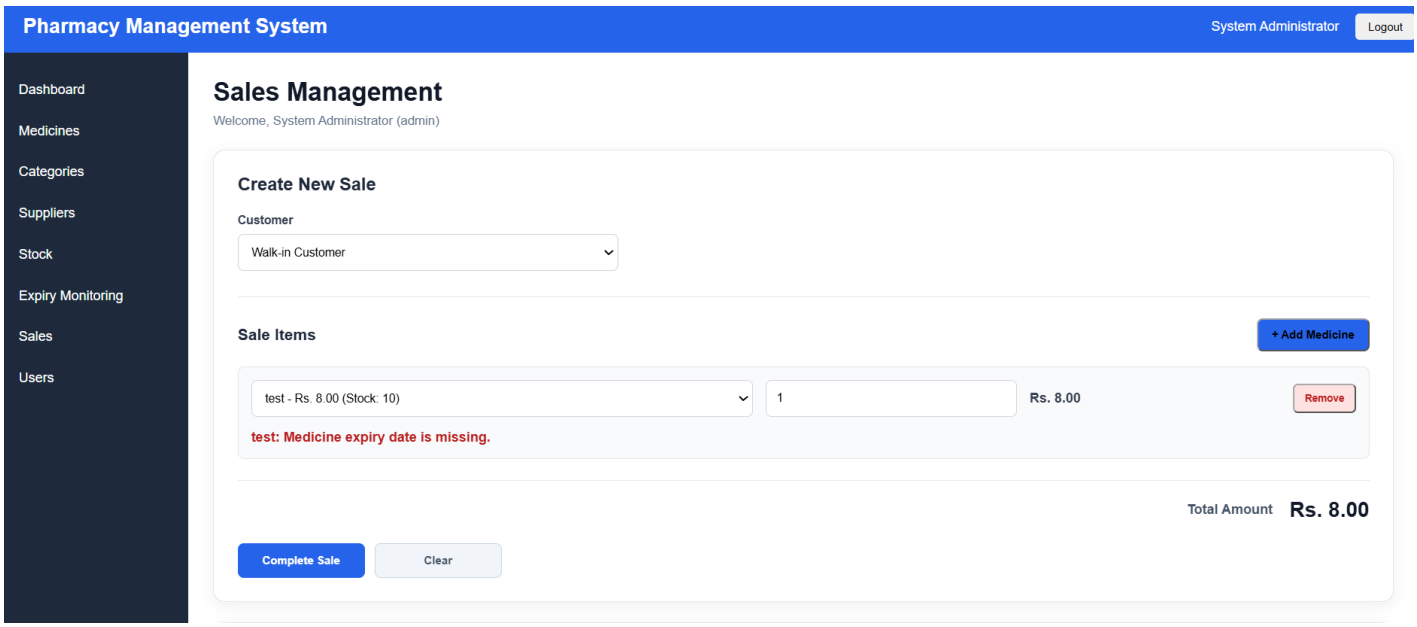
Evidence 01 ? Expired Medicine Rejection

. A screenshot from the running sales page was not supplied.



Evidence 02 ? Missing and Boundary Expiry Dates

confirmed rejection of missing expiry before transaction



s6. Final Result

Item	Result
Requirements	Implemented
Backend validation	Implemented
Frontend warning and submit prevention	Implemented
Testing	Implemented
Acceptance criteria	Implemented
Final Status	Implemented

6.1 Final Summary

CR-002 strengthens expiry validation in the sales workflow. Expired, missing-expiry, and invalid-expiry medicines are blocked by frontend checks and independently rejected by the backend before sale transaction or stock update. Targeted mocked-route checks passed for expired, missing, and today-expiry cases. Release sign-off requires browser and test-database verification, including mixed-item rejection and normal sales regression tests.

6.2 Issues / Limitations

Browser screenshots and live MySQL integration results were not available. Verify DATE_FORMAT behavior against the deployment database and run pending cases in a test environment before production sign-off.

Student Name:_____
Student ID: _____
Date: _____
Signature: _____